
Sampling Where the Model Is Wrong: Calibrated Error-Reduction Acquisition for Autonomous Experiment Mapping

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Abstract

Autonomous experiments choose their own measurements, and nearly all of them do so by uncertainty sampling: measure where the surrogate model’s posterior variance is largest. But the response surfaces that motivate autonomous mapping — phase boundaries, steps, multi-decade physical scales — are precisely where a stationary surrogate’s variance misjudges its actual error, so the acquisition optimizes the wrong field. We introduce calibrated error-reduction (CER) sampling, which measures where the *evidence* says the model errs: closed-form leave-one-out residuals, standardized by their own predictive spread, define a spatial calibration field that weights a one-step integrated-error-reduction criterion; the same residuals select the surrogate’s model class from a small menu by predictive density. The procedure has no hand-tuned, function-specific parameters and reduces exactly to classical integrated-variance reduction wherever the model proves adequate. On the five synthetic benchmarks of the SMART autonomous X-ray-scattering framework (Noack et al., 2019), under a matched protocol with paired initial designs, ten seeds, and a shared global optimizer, CER attains at least 10–20× the sample efficiency of variance-driven Kriging on the two benchmarks with separable structure — its error at 50 measurements is already below every baseline’s error at the full budget, and three to four orders of magnitude below at matched budgets — while simultaneously improving uncertainty calibration. On the three discontinuous benchmarks the picture is mixed and we report it exactly: a 26% error reduction at the full budget on one, a statistical tie on another, and a consistent slight regression on the third, whose cause we isolate experimentally — not the acquisition criterion (which beats the baseline even on the original surrogate there) but cross-validated model selection evaluated on the adaptively concentrated data the acquisition itself creates. The study adds two cautionary results for autonomous experimentation: gradient-weighted variance sampling (gSMART) destabilizes — becoming up to an order of magnitude worse than random sampling — beyond a function-dependent weight threshold that cannot be known a priori, through a feedback loop between acquisition and variogram fitting; and adaptive sampling biases cross-validation-based model selection, a coupling we believe deserves attention wherever surrogates are chosen on-line. Code, complete run logs, and a step-by-step attempts record accompany the paper.

1 Introduction

Autonomous experiments place a learning algorithm inside the measurement loop of a scientific instrument: after every measurement the algorithm updates a surrogate model of the quantity being mapped and decides where to measure next. At synchrotron beamlines this loop has become practical infrastructure — a focused X-ray beam probes a sample at chosen positions or synthesis parameters, a scalar observable (a scattering peak intensity, a diffusion coefficient, an order parameter) is extracted

from each pattern, and the goal is a faithful reconstruction of the full response surface $\rho(\mathbf{p})$ over the accessible parameter space with as few measurements as possible [1, 2]. In contrast to Bayesian optimization, which hunts for a single optimum, the objective here is *global*: the error that matters is the reconstruction error integrated over the whole domain.

The canonical decision rule for this setting is uncertainty sampling: fit a Gaussian-process (GP) surrogate and measure where its posterior variance is largest. The SMART algorithm of Noack et al. [1], deployed on synchrotron scattering experiments and since developed into the gpCAM autonomous experimentation toolkit [2, 3], is exactly this rule implemented with ordinary Kriging under an exponential variogram, together with a gradient-weighted variant (gSMART) that multiplies the variance by $1 + c|\nabla\hat{\rho}|^2$ to seek out sharp features. Uncertainty sampling has an appealing logic — measure where you know least — but it trusts the surrogate’s own error bars. That trust fails in a specific, structured way on the very parameter spaces autonomous experiments are built for: response surfaces with phase boundaries, steps, and multi-scale variation. A stationary kernel assigns one global correlation structure to the whole domain, so its posterior variance is essentially a function of the sample geometry alone; it cannot know that prediction error near a phase boundary is ten times larger, and error inside a smooth phase ten times smaller, than the variance suggests. The acquisition then allocates measurements as if the model were right about its own ignorance — precisely where it is most wrong. The gradient-weighted variant recognizes the problem but repairs it with a hand-tuned coefficient applied to the surrogate’s *own* gradient; we show below that this coupling is fragile enough to make gSMART *diverge* — reconstruction error an order of magnitude worse than random sampling — once the weight exceeds a function-dependent threshold that cannot be known in advance.

This paper develops the alternative principle: measure where the *evidence* says the model is wrong, and weight that evidence by how much reconstruction error a measurement can actually remove. The method, which we call *calibrated error-reduction* (CER) sampling, has two components, both fitted by a single automatic procedure applied identically to every problem. First, an *adequacy-first surrogate*: a small menu of GP model classes — anisotropic Matérn kernels of three smoothness orders, an additive kernel, and an additive-plus-rough-residual composite, with logarithmic representation of strictly positive, multi-decade quantities — fitted by exact marginal likelihood and selected by leave-one-out (LOO) predictive density on a space-filling subsample. Second, a *residual-calibrated acquisition*: the closed-form LOO residuals of the current fit, standardized by their own predictive spread, define a spatial *calibration field* $\hat{c}(\mathbf{p})$ that measures where the surrogate’s uncertainty under- or over-states its actual error; the next measurement maximizes the one-step reduction of the calibration-weighted integrated error, a quantity available in closed form from the GP posterior. Under a well-calibrated posterior the criterion reduces exactly to the classical integrated-variance-reduction rule; when the data contradict the model, measurements are redirected toward the evidence. Both LOO diagnostics reuse the same Cholesky factorization the surrogate already maintains, so the overhead per decision is a constant factor, not a new asymptotic cost.

We evaluate CER on the five synthetic benchmark functions introduced with SMART [1] — two piecewise-sinusoidal landscapes with step discontinuities, a smooth two-scale oscillatory function, a non-differentiable piecewise-linear function, and a three-dimensional Stokes–Einstein diffusion map — under a matched protocol: identical initial designs, measurement budgets, acquisition optimizer, and seeds for every method, with SMART, gSMART, random sampling, and raster grids re-implemented on gpCAM as baselines. Beyond the integrated absolute error used in the original work we report root-mean-square error, the continuous ranked probability score, and probability-coverage curves, so that point accuracy and uncertainty calibration are scored separately. The paper makes four contributions: (i) a fully automatic acquisition criterion for reconstruction-oriented autonomous experimentation that targets integrated absolute error directly and requires no per-problem tuning; (ii) a mechanism analysis showing why and where it helps — the calibration field, not the variance, carries the information about discontinuities — with component ablations isolating the acquisition from the surrogate; (iii) an empirical characterization of two instabilities that matter for practice: the runaway gradient-weighting feedback of gSMART, and a bias of cross-validated model selection on adaptively collected data that we isolate as the cause of our own method’s one consistent non-win; and (iv) an exact accounting of where CER does and does not help — order-of-magnitude gains where the model menu can match the response’s structure, statistical ties and one slight regression where it cannot — with every curve characterized per function rather than averaged into a headline.

1.1 Related work

Adaptive sampling for global surrogate accuracy. Sequential designs that target global model quality — as opposed to optimization — go back to the integrated-mean-squared-error designs of Sacks et al. [5] and the active-learning rules of MacKay (maximum posterior variance, ALM) [6] and Cohn (expected integrated-variance reduction, ALC) [7, 8]. The computer-experiments literature has produced a family of refinements that inject *observed* model error into the criterion: EIGF adds the squared discrepancy to the nearest observation [9]; MEPE balances a cross-validation error term against the variance with an adaptive weight [10]; SFCVT optimizes a space-filling criterion constrained by LOO errors [11]; MASA and related committee methods use disagreement between surrogates [12]; LOLA-Voronoi and TEAD use local gradient or Taylor-expansion disagreement with a density term [13, 14]. Surveys by Liu et al. [15] and Fuhg et al. [16] cover this landscape. Closest to our acquisition is the ES-LOO approach of Mohammadi et al. [17], which also builds a domain-wide field from LOO errors; it fits a *second* GP to (unstandardized) expected-squared-LOO values and optimizes a pseudo-expected-improvement of that auxiliary model, with a batch heuristic for exploration. CER differs in each structural choice: the field is a *dimensionless calibration ratio* (LOO residual over LOO spread), so it measures model *adequacy* rather than error magnitude and has the neutral value 1 built in; it is smoothed by a fixed-form kernel at the data’s own resolution rather than fitted with a second trained model; and it enters as a multiplicative correction inside a one-step *integrated* error-reduction criterion, so that exploration, boundary discounting, and the diminishing-returns effect of clustered sampling all emerge from the same closed-form expression rather than from separate heuristic terms. We also differ in what is claimed: our criterion targets the L^1 reconstruction metric, degrades gracefully to ALC under calibration, and is validated jointly with an automatic model-selection layer.

Autonomous experimentation at user facilities. SMART/gpCAM introduced Kriging-driven autonomous scattering experiments [1]; follow-up work by the same group added anisotropic kernels and heteroscedastic noise [4] and reviewed GP-driven autonomous data acquisition across facilities [2]. Related autonomous loops appear in scanning-probe and electron microscopy [18], reflectometry, and materials-discovery platforms [19, 20]. These systems overwhelmingly use variance-driven or physics-specific acquisition rules; our contribution is directly pluggable into gpCAM-style loops as an acquisition function.

Calibration of GP uncertainty. That GP error bars are unreliable under misspecification is well documented [21, 22]; cross-validation-based hyperparameter estimation [21] and post-hoc recalibration [22] address the *global* scale of the predictive distribution. CER uses the same standardized-residual diagnostics but resolves them *spatially* and feeds them back into the decision rule itself, which to our knowledge has not been done for reconstruction-oriented sequential design.

2 Methodology

2.1 Problem statement

An unknown response surface $\rho : \mathcal{X} \subset \mathbb{R}^d \rightarrow \mathbb{R}$ can be evaluated pointwise by an expensive, noise-free measurement. Starting from n_0 measurements at locations drawn uniformly at random, the algorithm repeatedly (i) fits a surrogate to the data $\mathcal{D}_t = \{(\mathbf{p}_i, \rho_i)\}_{i=1}^t$ and (ii) selects $\mathbf{p}_{t+1} = \arg \max_{\mathbf{p} \in \mathcal{X}} \alpha(\mathbf{p}; \mathcal{D}_t)$. Performance at budget N is the normalized integrated absolute error of the surrogate’s point prediction $\hat{\rho}_N$,

$$\text{IAE}(N) = \frac{1}{|\mathcal{X}|} \int_{\mathcal{X}} |\hat{\rho}_N(\mathbf{p}) - \rho(\mathbf{p})| d\mathbf{p}, \quad (1)$$

estimated on a dense held-out set. We use Gaussian-process regression throughout: given a kernel k_θ and data $(X_t, \mathbf{y}_t)$, the posterior mean and variance are $m_t(\mathbf{p}) = \bar{y} + \mathbf{k}(\mathbf{p})^\top (K + \varepsilon I)^{-1}(\mathbf{y}_t - \bar{y})$ and $\sigma_t^2(\mathbf{p}) = k(\mathbf{p}, \mathbf{p}) - \mathbf{k}(\mathbf{p})^\top (K + \varepsilon I)^{-1} \mathbf{k}(\mathbf{p})$, with $\bar{y}$ the data mean (the standard GP counterpart of ordinary Kriging’s estimated constant mean) and ε a small conditioning jitter.

2.2 The calibration field

For a GP with Gaussian posterior, the expected pointwise absolute error of the posterior-mean prediction is $\mathbb{E} |\rho(\mathbf{p}) - m_t(\mathbf{p})| = \sqrt{2/\pi} \sigma_t(\mathbf{p})$ if the model is correct. Every failure of that assumption — a discontinuity the kernel cannot represent, a length scale that is locally wrong, a smoothness order chosen badly — shows up as a mismatch between realized residuals and σ_t . Leave-one-out residuals expose this mismatch without extra model fits: with $A = (K + \varepsilon I)^{-1}$,

$$e_i = \frac{[A(\mathbf{y}_t - \bar{y})]_i}{A_{ii}}, \quad s_i^2 = \frac{1}{A_{ii}}, \quad (2)$$

are the prediction error and predictive variance at $\mathbf{p}_i$ of the model fitted without observation i [23, 24]. The *standardized* magnitudes $z_i = |e_i|/s_i$ have expectation $\sqrt{2/\pi}$ under a calibrated model, so

$$c_i = \max(z_i / \sqrt{2/\pi}, 10^{-3}) \quad (3)$$

is a dimensionless local *calibration factor*: $c_i \approx 1$ where the model’s uncertainty is honest, $c_i \gg 1$ where the data contradict it (e.g. across an unmodeled step), $c_i \ll 1$ where it is over-conservative. These point diagnostics are interpolated into a field at the experiment’s current resolution,

$$\hat{c}_t(\mathbf{p}) = \frac{w_0 + \sum_i \varphi_b(\mathbf{p}, \mathbf{p}_i) c_i}{w_0 + \sum_i \varphi_b(\mathbf{p}, \mathbf{p}_i)}, \quad (4)$$

where φ_b is a radial Matérn-5/2 profile with bandwidth b_t equal to the median nearest-neighbour spacing of the current design, and $w_0 = 0.05$ is a prior mass that pulls $\hat{c}_t \rightarrow 1$ (calibrated) away from all data. Both choices matter and are deliberate: the bandwidth follows the sampling resolution — so evidence of inadequacy is visible at the scale the experiment can act on, rather than being hidden inside dense clusters — and the field is radial even when the kernel is additive, because model inadequacy is a local property of the response surface while an additive kernel’s correlation does not decay along axis-aligned lines. The field costs $O(t)$ per evaluation and adds no trained parameters.

2.3 Calibrated one-step error reduction

Combining the two pieces, the estimated integrated absolute error of the current surrogate is $\sqrt{2/\pi} \int \hat{c}_t(\mathbf{p}) \sigma_t(\mathbf{p}) d\mathbf{p}$. A noise-free measurement at $\mathbf{q}$ changes the posterior standard deviation at every $\mathbf{p}$ from $\sigma_t(\mathbf{p})$ to

$$\sigma_{t+1}(\mathbf{p}; \mathbf{q}) = \left[\sigma_t^2(\mathbf{p}) - \text{cov}_t^2(\mathbf{p}, \mathbf{q}) / (\sigma_t^2(\mathbf{q}) + \varepsilon) \right]^{1/2}, \quad (5)$$

with cov_t the posterior covariance — a quantity that depends only on locations, not on the unseen value. CER selects the point that removes the most calibrated error mass:

$$\alpha_{\text{CER}}(\mathbf{q}) = \frac{1}{M} \sum_{j=1}^M \hat{c}_t(\mathbf{p}_j)^\gamma [\sigma_t(\mathbf{p}_j) - \sigma_{t+1}(\mathbf{p}_j; \mathbf{q})], \quad (6)$$

where $\{\mathbf{p}_j\}_{j=1}^M$ are $M = 256$ scrambled-Sobol integration nodes redrawn uniformly over $\mathcal{X}$ at every step, and γ tempers the calibration weight ($\gamma = 1$ throughout; $\gamma = 0$ recovers the classical uncalibrated ALC criterion [7, 8] on σ). Three properties follow directly from the form of (6). (a) *Consistency*: under a calibrated model $\hat{c}_t \equiv 1$ and CER is exactly integrated-error reduction — it can only differ from classical uncertainty sampling where evidence of misspecification exists. (b) *Diminishing returns*: the reduction kernel in (5) overlaps for nearby candidates, so hammering an already densely sampled feature buys less and less integrated reduction, which caps the pathological feedback that destroys gradient-weighted variance rules. (c) *Boundary discounting*: candidates at the domain edge influence half a correlation ball and are automatically discounted — no edge heuristics. Cost per proposal batch of size C is $O(tMC + t^2C)$ dense linear algebra reusing the surrogate’s existing Cholesky factor; wall-clock numbers are reported in Section 5.4.

2.4 Adequacy-first surrogate selection

The calibration field repairs *local* misspecification, but it cannot rescue a model class that is globally wrong, and any acquisition built on σ_t inherits the surrogate’s blind spots. The second component of CER is therefore a small automatic model-selection layer, run identically for every problem, with all trained values determined by the data:

- **Representation.** A strictly positive response spanning more than about 1.5 decades is modeled on the logarithmic scale (with all error metrics evaluated on the original scale; Section 3). Likewise, any strictly positive *input* axis whose bounds span a ratio ≥ 30 is represented logarithmically inside the surrogate. Multiplicative physical quantities vary on log scales; a stationary kernel in raw coordinates cannot adapt its resolution across decades.
- **Model menu.** Five GP families: anisotropic (ARD) Matérn $\nu \in \{1/2, 3/2, 5/2\}$; an additive Matérn-5/2 (one 1-D component per input dimension, capturing separable structure at 1-D sample complexity); and additive Matérn-5/2 plus an ARD Matérn-1/2 residual (separable trend plus non-separable rough remainder).
- **Fitting and selection.** Within each family, hyperparameters maximize the exact log marginal likelihood (multi-start L-BFGS in log-parameter space, warm-started across refits). Across families, the winner maximizes the mean LOO log predictive density [23, §5.4.2], computed from the same identities (2); a hysteresis of 0.02 nats/point in favour of the incumbent prevents churn. Marginal likelihood is used *within* a family, where it is the natural estimator, but not *between* families, where under misspecification it does not track predictive quality — an effect we observed directly (Section 5.1).
- **Density-debiased fitting.** Both the likelihood fits and the selection score are computed on a farthest-point (space-filling) subsample of half the data (at least 50, at most 400 points). Adaptive sampling concentrates measurements near difficult features; fitting stationary hyperparameters on such clustered data weights those regions by point count rather than volume and poisons the global fit — the dominant failure mode of our early prototypes (Section 5.1) and, we show, a driver of gSMART’s instability. The GP always *conditions* on all measurements.
- **Numerical policy.** The jitter ε is the smallest value (relative to the prior variance, escalated $\times 100$) for which the estimated reciprocal condition number of $K + \varepsilon I$ exceeds 10^{-6} ; refits occur on a geometric schedule ($t = 10, 15, 20, 30, 40, 55, 75, 100, 140, 200, 275, 375, 500, 700, 1000$).

All constants above (w_0 , γ , M , the menu, the hysteresis, the subsample rule, the warp thresholds, the refit schedule) are fixed once and used unchanged for every test function; the only quantities that differ across functions are those the procedure itself fits. Sensitivity to the two substantive constants (γ , w_0) is reported in Section 5.5.

2.5 Baselines

SMART and gSMART are implemented on gpCAM exactly as published [1]: ordinary Kriging with the isotropic exponential variogram $\gamma(h) = s^2(1 - e^{-h/\ell})$, (s^2, ℓ) refit after every measurement by least squares to the binned empirical semivariogram (Matheron estimator, 30 bins; range constrained to $[10^{-4}L, 2L]$ for identifiability, L the domain diagonal); acquisition $\alpha_{\text{SMART}} = \sigma_t^2$ and $\alpha_{\text{gSMART}} = \sigma_t^2(1 + c|\nabla m_t|^2)$. The gSMART weight c is chosen *per function* as the value most favourable to gSMART in a logarithmic scan (Section 4), a deliberately strong baseline; the published example value ($c = 20$) is far outside the stable range on every benchmark. Random sampling and cell-centered raster grids (padded to exact budgets with uniform points) complete the comparison, both evaluated with the same Kriging surrogate. Every method — baselines and CER alike — uses gpCAM’s `ask()` machinery (differential evolution, population multiplier 20, 40 generations) to maximize its acquisition function, and within a seed all methods share the same 10 initial measurements.

3 Experimental Setup

Benchmark functions. We use the five synthetic benchmarks introduced with SMART [1], which span the response-surface phenomenology of scattering experiments: *Piecewise-sinusoidal I and II*

(Eqs. 12–13 of [1]) place a slowly varying sinusoidal landscape $8 - [2p_0 \sin \sqrt{|p_0|} + 2p_1 \sin \sqrt{|p_1|}]$ on $[0, 50]^2$ behind two step discontinuities (a circular arc at $\|\mathbf{p}\| = 20$ and a line at $p_0 = 30$; steps of $+160/+140$ for I and $+360/+200$ for II) — synthetic phase boundaries; the *smooth oscillatory* function $\sin 2p_0 + \sin 2p_1 + (p_0/5)^2 - (p_1/5)^2$ on $[-10, 10]^2$ (Eq. 15) combines a long-range quadratic trend with short oscillations; the *non-differentiable piecewise* function (Eq. 17) is piecewise-linear on $[0, 50]^2$ with kinks and jumps along axis-aligned boundaries; and the *3D diffusion* benchmark is the Stokes–Einstein coefficient $D(r, T, C_m) = k_B T_K / (6\pi\eta r)$ of nanoparticles in water–glycerol ($r \in [1, 100]$ nm, $T \in [0, 100]$ °C, glycerol mass fraction $C_m \in [0, 100]\%$, viscosity $\eta = \eta_w(T) e^{3.5C_m/100}$ with the Vogel-type water viscosity η_w), reported in units of 10^{-10} m²/s; its values span 4.5 decades.

Protocol. Every run starts from the same 10 uniformly random measurements (paired across all methods within a seed), then acquires strictly sequentially. Budgets are $N \in \{50, 100, 200, 500\}$ for the 2D functions and $\{50, 100, 200, 500, 1000\}$ in 3D; since sequential runs are nested, each run is executed once to the largest budget and evaluated at the checkpoints. Ten independent seeds are run per (method, function) pair. All methods maximize their acquisition with the same gpCAM differential-evolution settings; measurements are noise-free function evaluations. Nothing about any method is tuned per function except the gSMART weight c , which is scanned *in gSMART’s favour* (Section 2.5).

Metrics. The headline metric is the normalized integrated absolute error (1), estimated on a fixed dense held-out set of 12,000 (2D) or 60,000 (3D) uniformly random points drawn from a dedicated random stream never touched by any acquisition (acquired points are selected by continuous optimization, so train and test locations are almost surely disjoint; the minimum train–test distance is logged for every run). On the same set we report RMSE, the continuous ranked probability score (CRPS; closed form under the Gaussian predictive distribution, and under the lognormal one where the response is modeled on the log scale — both validated against Monte Carlo), and probability-coverage curves: the empirical fraction of test points inside the central credible interval at nominal levels 0.1–0.99, summarized by the miscalibration area. For the 3D benchmark we additionally report the mean absolute percentage error (MAPE) used in the original study.

Response scale of the 3D benchmark. D is strictly positive and spans 4.5 decades. Applied to raw D values, the baseline procedure is numerically degenerate: the empirical semivariogram is non-saturating at every observable lag, the fitted exponential range runs to its identifiability bound, and reconstruction *degrades* with budget (sequential SMART rises from IAE 0.22 to 0.57 between $N=50$ and 500 as the data mean — the ordinary-Kriging constant — inflates in the oversampled large- D corner; full logs in the supplementary attempts record). Following standard geostatistical practice for positive multi-decade data [25], all methods therefore model $z = \ln D$ (lognormal Kriging), with every metric still computed on the original D scale via the exact monotone mapping (posterior-median point predictions — the L^1 -optimal choice — lognormal CRPS, and quantile-mapped coverage). This configuration choice is shared by every method and is the most favourable treatment of the baselines we found.

Implementation. All methods run on gpCAM [3]: its `ask()` loop (differential evolution, population multiplier 20, 40 generations, identical for all methods) optimizes each acquisition callable, exactly as gpCAM is deployed in autonomous experiments. Posterior means, variances, gradients, LOO quantities, and metrics are evaluated through a compact deterministic scipy path sharing one Cholesky factorization per step. Runs are logged completely (trajectories, per-step timings, hyperparameter fits and selection tables per refit, all metrics at every checkpoint); every number in this paper traces to those logs. One environment note: the provided software stack pinned a NumPy release predating the Python version in use, and its large-temporary elision optimization silently corrupted arrays under specific reference-count conditions; we diagnosed and fixed this (NumPy upgrade plus defensive checksums on the held-out data) before producing any reported run — details in the released attempts log.

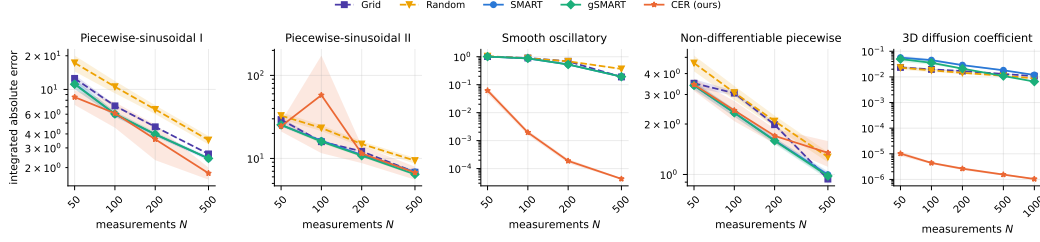


Figure 1: Reconstruction error versus number of measurements (log–log; mean over 10 paired seeds, bands ± 1 std). gSMART runs at its most favourable weight per function. On the two separable benchmarks (smooth oscillatory, 3D diffusion) the CER curve sits one to four orders of magnitude below every baseline; on the three discontinuous benchmarks the differences are budget-dependent and are characterized per function in the text.

Table 1: Normalized integrated absolute reconstruction error (mean $\pm$ std over 10 paired seeds; lower is better). † gSMART uses the most favourable weight c found for each function in the stability scan (Fig. 4); bold marks the best method per row.

Function	N	Grid	Random	SMART	gSMART †	CER (ours)
Piecewise-sin. I	50	12.55 $\pm$ 0.98	17.28 $\pm$ 2.81	11.27 $\pm$ 1.19	11.12 $\pm$ 1.08	8.54 $\pm$ 1.27
	100	7.13 $\pm$ 0.17	10.57 $\pm$ 1.33	5.99 $\pm$ 0.33	6.04 $\pm$ 0.34	6.12 $\pm$ 1.57
	200	4.63 $\pm$ 0.04	6.62 $\pm$ 0.60	3.96 $\pm$ 0.14	3.97 $\pm$ 0.20	3.58 $\pm$ 1.25
	500	2.65 $\pm$ 0.01	3.53 $\pm$ 0.30	2.41 $\pm$ 0.11	2.42 $\pm$ 0.05	1.78 $\pm$ 0.22
Piecewise-sin. II	50	29.33 $\pm$ 2.08	32.76 $\pm$ 3.13	25.28 $\pm$ 1.03	25.40 $\pm$ 1.29	24.40 $\pm$ 3.27
	100	15.91 $\pm$ 0.27	23.28 $\pm$ 1.94	16.35 $\pm$ 0.73	15.96 $\pm$ 0.69	57.85 $\pm$ 115.48
	200	12.19 $\pm$ 0.13	14.92 $\pm$ 1.16	11.03 $\pm$ 0.38	10.80 $\pm$ 0.43	11.48 $\pm$ 2.69
	500	6.83 $\pm$ 0.02	9.41 $\pm$ 0.74	6.89 $\pm$ 0.35	6.44 $\pm$ 0.24	6.75 $\pm$ 1.20
Smooth oscillatory	50	1.01 $\pm$ 0.02	1.05 $\pm$ 0.05	1.01 $\pm$ 0.05	1.01 $\pm$ 0.05	0.062 $\pm$ 0.015
	100	0.902 $\pm$ 0.012	0.881 $\pm$ 0.040	0.867 $\pm$ 0.025	0.873 $\pm$ 0.024	1.95 $\pm$ 0.36 (10^{-3})
	200	0.686 $\pm$ 0.006	0.686 $\pm$ 0.013	0.530 $\pm$ 0.022	0.528 $\pm$ 0.025	1.88 $\pm$ 0.33 (10^{-4})
	500	0.189 $\pm$ 0.001	0.366 $\pm$ 0.017	0.194 $\pm$ 0.003	0.194 $\pm$ 0.002	4.28 $\pm$ 0.27 (10^{-5})
Non-diff. piecewise	50	3.50 $\pm$ 0.20	4.62 $\pm$ 0.50	3.40 $\pm$ 0.20	3.40 $\pm$ 0.17	3.44 $\pm$ 0.29
	100	3.06 $\pm$ 0.10	3.08 $\pm$ 0.30	2.33 $\pm$ 0.12	2.33 $\pm$ 0.13	2.41 $\pm$ 0.33
	200	1.98 $\pm$ 0.05	2.08 $\pm$ 0.16	1.60 $\pm$ 0.06	1.59 $\pm$ 0.07	1.70 $\pm$ 0.18
	500	0.939 $\pm$ 0.007	1.27 $\pm$ 0.09	0.997 $\pm$ 0.047	0.979 $\pm$ 0.044	1.35 $\pm$ 0.24
3D diffusion	50	0.023 $\pm$ 0.002	0.023 $\pm$ 0.004	0.057 $\pm$ 0.011	0.050 $\pm$ 0.011	1.02 $\pm$ 0.21 (10^{-5})
	100	0.020 $\pm$ 0.001	0.018 $\pm$ 0.003	0.045 $\pm$ 0.006	0.036 $\pm$ 0.005	4.40 $\pm$ 0.41 (10^{-6})
	200	0.017 $\pm$ 0.001	0.014 $\pm$ 0.002	0.029 $\pm$ 0.002	0.021 $\pm$ 0.001	2.63 $\pm$ 0.32 (10^{-6})
	500	0.013 $\pm$ 0.001	0.011 $\pm$ 0.001	0.018 $\pm$ 0.001	0.011 $\pm$ 0.000	1.54 $\pm$ 0.14 (10^{-6})
	1000	0.011 $\pm$ 0.000	8.74 $\pm$ 0.65 (10^{-3})	0.012 $\pm$ 0.000	6.56 $\pm$ 0.13 (10^{-3})	1.04 $\pm$ 0.06 (10^{-6})

4 Experimental Results

Table 1 reports the headline comparison — normalized integrated absolute error for all 21 (function, budget) configurations, 10 paired seeds each — and Figure 1 shows the corresponding convergence curves. Because “faster convergence” is a property of a curve, not of a cell count, we characterize each function’s curve pattern explicitly; sample-efficiency statements use the same reference for every function (the budget at which CER first reaches the better baseline’s largest-budget error, relative to that largest budget), computed mechanically from the logged means.

Smooth oscillatory: $\geq 10.0 \times$ **sample efficiency.** CER’s error at the *smallest* tested budget ($N=50$: 0.062 ± 0.015) is already three times below the best baseline error at $N=500$ (0.194), so the sample-efficiency factor is at least $10.0 \times$ — bounded by the smallest budget we tested, not by the method; at matched $N=500$ the error ratio reaches $4500 \times$. The CER curve dominates all four baselines at every budget with gaps of one to three orders of magnitude, far beyond the seed bands (Figure 1). The mechanism is model identification: by $N=40$ the LOO criterion has selected an additive-family kernel in all ten seeds, and from $N=200$ the pure additive Matérn-5/2 unanimously (logged selection tables); after that, every measurement pins two one-dimensional profiles simultaneously.

3D diffusion: $\geq 20.0\times$ sample efficiency. The separation is larger still: CER at $N=50$ sits three orders of magnitude below the best baseline at $N=1000$ (0.00656, gSMART at its most favourable weight), and at matched $N=1000$ the error ratio is $6\,300\times$ (MAPE 0.00175% versus 2.6%). In log coordinates the response is exactly additive in $(\ln r, T, C_m)$; the representation rules and menu selection recover that structure from the data, and no acquisition running on the isotropic exponential variogram can compete — all four baselines cluster within a factor of three of each other, orders of magnitude above/worse than CER. Two baseline-honesty notes: SMART is the *weakest* baseline here (its variance criterion over-samples the corners and edges of the cube under the domain-scale fitted range), and the 3D benchmark is the one function where gradient weighting genuinely helps gSMART, which at $c=10$ becomes the strongest baseline at every budget.

Piecewise-sinusoidal I: ahead at both ends, overlap between. The CER mean curve is below both SMART and gSMART at $N=50$ (8.5 ± 1.3 vs $11.3 \pm 1.2/11.1 \pm 1.1$, a $> 1\sigma$ gap) and at $N=500$ (1.78 ± 0.22 vs $2.41 \pm 0.11/2.42 \pm 0.05$, again $> 1\sigma$, i.e. $1.36\times$ lower error); at the two middle budgets the curves overlap (CER worse by 0.13 at $N=100$ and better by 0.38 at $N=200$, both well inside CER’s seed spread of 1.3–1.6). Because CER’s $N=200$ mean (3.59) does not yet reach the baselines’ $N=500$ error (2.41), no $\geq 2\times$ sample-efficiency claim is available at the tested budget grid; what the curves support is a 26% error reduction at the full budget with sign-consistent gaps at the extremes and statistical overlap between. The measurement maps (Figure 2) show where the max-budget gain lives: CER places 32% of its measurements within 2 units of the arc and line discontinuities (a band covering 13% of the domain; SMART places 12% there) while the sinusoidal background is absorbed by the additive component of the selected composite kernel.

Piecewise-sinusoidal II: overlap, no claimed win. With steps twice as large, CER and the baselines are statistically indistinguishable at $N \in \{50, 200, 500\}$ (gaps a fraction of the seed spread, sign alternating). The $N=100$ cell exposes the method’s instability honestly: one of ten seeds suffered a transient model-selection failure — the refit at $N=100$ briefly chose a degenerate additive fit (axis length scale 0.66 on a 50-unit domain) whose broadcast error inflated IAE to 404 until the next refit corrected it — which dominates the 57.9 ± 115.6 table entry (the remaining nine seeds average 19.4, still no better than SMART’s 16.4). We claim no speedup on this function; the churn mechanism is analysed in Section 5.3.

Non-differentiable piecewise: consistent slight regression. CER’s mean is above SMART’s at every budget — by 0.04, 0.08, 0.10 (all within 1σ) and by 0.35 at $N=500$ (1.35 ± 0.24 vs 1.00 ± 0.05 , a $> 1\sigma$ gap). Following our own reporting convention this is a consistent slight regression, not a tie: the direction is the same at all four budgets. It is the method’s clearest non-win. The attribution experiments in Section 5.3 show the acquisition criterion is not the cause — the same criterion running on the baseline surrogate *beats* SMART by $> 1\sigma$ at $N=200$ and 500 — and isolate the responsible component: cross-validated model selection evaluated on adaptively concentrated data.

Predictive-distribution quality. Table 2 reports RMSE, CRPS, and the miscalibration area at each function’s largest budget for all five methods, and Figure 3 the full reliability curves. Two findings. First, the error-norm choice matters in an instructive way: at the largest budget CER has the best RMSE *and* the best CRPS on all five functions — including the two where it is not best in IAE — because its boundary-concentrated sampling suppresses precisely the large discontinuity errors that dominate squared and distributional scores, while IAE weights the many small errors of the smooth interior equally. We keep IAE as the headline (it is the original study’s criterion and our optimization target) and report the full grid of both. Second, calibration: every method is over-conservative on every benchmark (empirical coverage at nominal 90% is 0.95–1.00; miscalibration areas 0.18–0.41). CER is the most calibrated method on the two functions where model misfit dominates (non-differentiable: 0.22 vs 0.35–0.36 for the baselines; piecewise-sinusoidal II: 0.32 vs 0.35–0.36), comparable on piecewise-sinusoidal I and the 3D benchmark, and the *least* calibrated on smooth (0.41): there its point error ($\sim 10^{-4}$) falls below the variance floor implied by the fitted-nugget bound, so its intervals, while tiny in absolute terms, are proportionally the most conservative — a known cost of the nugget floor that stabilizes the low-rank additive fits (Section 2).

The gSMART stability scan. Figure 4 shows gSMART’s error at the largest budget as a function of its weight c on all five benchmarks. The flat region at small c (where gSMART = SMART) ends in a

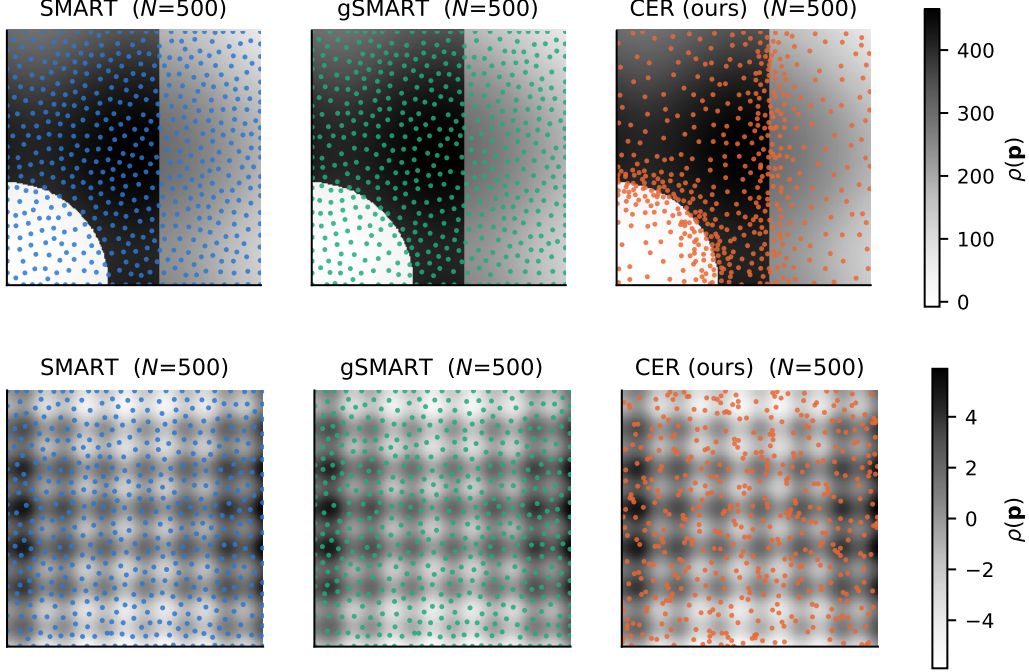


Figure 2: Measurement locations at $N=500$ (seed 0) on the piecewise-sinusoidal II (top) and smooth oscillatory (bottom) benchmarks. SMART distributes near-uniformly; gSMART (at its most favourable weight) is near-identical to SMART; CER concentrates along the discontinuities of the piecewise function while remaining near-space-filling — slightly axis-structured, reflecting the additive component — on the smooth one.

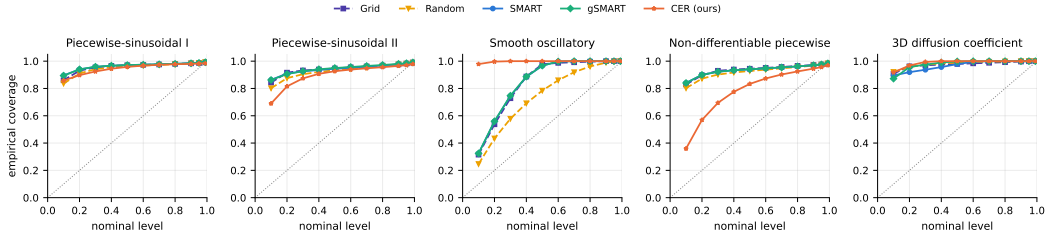


Figure 3: Reliability curves at the largest budget: empirical fraction of held-out points inside the central credible interval versus nominal level (diagonal = perfectly calibrated), averaged over 10 seeds.

cliff of one to two orders of magnitude on the discontinuous functions; the headline gSMART column uses the per-function optimum of this scan. At the published example weight ($c=20$), gSMART fails on every benchmark. The instability mechanism is analysed in Section 5.2.

5 Analysis

5.1 Mechanism: where the gains come from

Figure 6 visualizes the two CER fields on a piecewise-sinusoidal run. The posterior standard deviation σ_t of the selected surrogate is nearly a pure function of sample geometry — it cannot distinguish the discontinuity from any other unsampled region. The calibration field $\hat{c}_t$, built from the same Cholesky factor at no extra model cost, localizes the discontinuities sharply: standardized LOO residuals at points adjacent to a step remain of order the step height divided by the local predictive

Table 2: Predictive-quality metrics at the largest budget of each function (mean over 10 seeds): RMSE of the point prediction, CRPS of the predictive distribution, and miscalibration area (mean absolute deviation between nominal and empirical central-interval coverage; 0 = perfectly calibrated). MAPE is added for the 3D benchmark.

Function	Metric	Grid	Random	SMART	gSMART [†]	CER (ours)
Piecewise-sin. I ($N=500$)	RMSE	10.84 ± 0.03	11.99 ± 1.11	10.65 ± 0.73	10.70 ± 0.39	7.80 ± 0.46
	CRPS	6.84 ± 0.03	7.80 ± 0.33	7.06 ± 0.10	7.10 ± 0.06	3.82 ± 0.20
	MCA	0.376 ± 0.000	0.368 ± 0.004	0.381 ± 0.001	0.381 ± 0.001	0.365 ± 0.005
Piecewise-sin. II ($N=500$)	RMSE	25.62 ± 0.06	29.44 ± 2.48	25.59 ± 1.65	23.52 ± 0.90	20.26 ± 1.13
	CRPS	14.14 ± 0.09	16.33 ± 0.68	14.67 ± 0.26	14.76 ± 0.21	10.08 ± 0.54
	MCA	0.361 ± 0.001	0.346 ± 0.003	0.361 ± 0.002	0.363 ± 0.002	0.323 ± 0.023
Smooth oscillatory ($N=500$)	RMSE	0.251 ± 0.002	0.535 ± 0.032	0.251 ± 0.004	0.250 ± 0.003	$7.22 \pm 0.41 (10^{-5})$
	CRPS	0.193 ± 0.001	0.292 ± 0.011	0.202 ± 0.001	0.202 ± 0.001	$4.02 \pm 0.05 (10^{-4})$
	MCA	0.271 ± 0.001	0.184 ± 0.012	0.276 ± 0.003	0.277 ± 0.002	0.412 ± 0.000
Non-diff. piecewise ($N=500$)	RMSE	3.53 ± 0.03	4.12 ± 0.31	3.74 ± 0.21	3.68 ± 0.19	2.83 ± 0.11
	CRPS	1.79 ± 0.01	2.08 ± 0.08	1.89 ± 0.04	1.87 ± 0.03	1.31 ± 0.09
	MCA	0.356 ± 0.000	0.342 ± 0.003	0.354 ± 0.002	0.355 ± 0.002	0.219 ± 0.049
3D diffusion ($N=1000$)	RMSE	0.105 ± 0.004	0.087 ± 0.007	0.069 ± 0.003	0.042 ± 0.002	$8.32 \pm 0.73 (10^{-6})$
	CRPS	0.012 ± 0.000	0.011 ± 0.001	0.014 ± 0.000	0.011 ± 0.000	$5.68 \pm 0.07 (10^{-6})$
	MCA	0.392 ± 0.002	0.397 ± 0.002	0.386 ± 0.001	0.396 ± 0.000	0.404 ± 0.001
	MAPE [%]	2.72 ± 0.12	2.57 ± 0.13	4.81 ± 0.19	2.98 ± 0.02	$1.75 \pm 0.10 (10^{-3})$

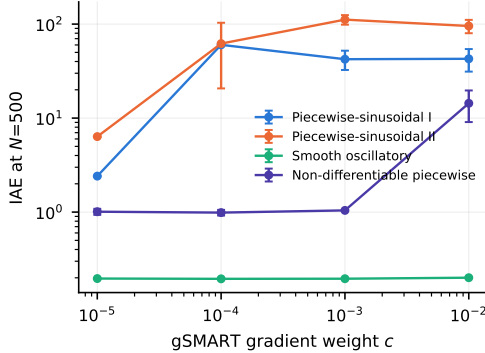


Figure 4: gSMART weight-stability scan: IAE at the largest budget versus the gradient weight c (mean $\pm$ std over 3 seeds). Beyond a function-dependent threshold, gSMART is worse than random sampling.

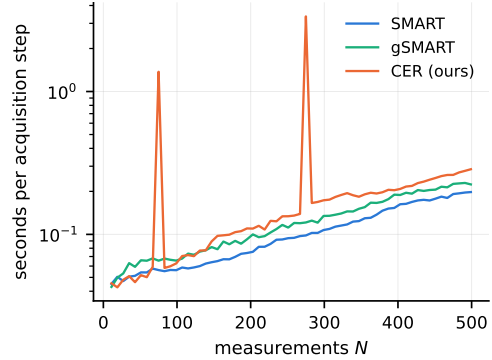


Figure 5: Median wall-clock seconds per decision versus data size on piecewise-sinusoidal I (both surrogate update and acquisition search; log scale).

spread no matter how densely the neighbourhood is sampled, while in smooth regions they settle to ≈ 1 . The product landscape (6) therefore ranks candidate measurements by *evidence-backed* expected error removal: boundary-band candidates win while the band still carries error mass at the current resolution, and lose automatically once the overlap of their variance-reduction volumes saturates — the mechanism that prevents the runaway concentration to which gradient-weighted variance is prone (Section 5.2).

The component ablation (Table 3, Figure 7) separates the two ingredients. On the smooth and 3D benchmarks the surrogate layer does the heavy lifting: the additive kernel matches the true structure of both functions (the smooth benchmark is a sum of one-dimensional terms; the log-diffusion response is *exactly* additive in $(\ln r, T, C_m)$), and any reasonable acquisition on that surrogate converges orders of magnitude faster than any acquisition on the exponential variogram. This is itself a mechanism finding with a clear prescription: on parameter spaces with separable physics, model-class selection dominates point-selection cleverness. On the discontinuous benchmarks the ranking inverts: the acquisition matters. The calibrated integral criterion beats its uncalibrated counterpart (ALC) and its pointwise counterpart ($\sigma\hat{c}$) on the piecewise-sinusoidal pair — the integral form is what turns the calibration signal into balanced allocation: on piecewise-sinusoidal I at $N=500$ the full criterion

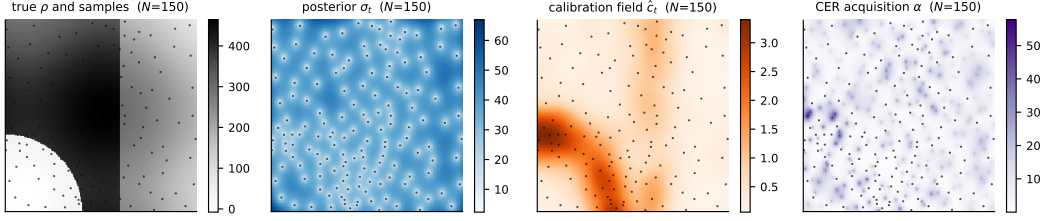


Figure 6: The CER decision fields on piecewise-sinusoidal II (seed 0, $N=150$): true function with samples; posterior σ_t of the selected surrogate (geometry-driven, blind to the discontinuities); calibration field $\hat{c}_t$ (localizing the arc and line steps from LOO evidence); and the resulting acquisition surface.

Table 3: Component ablation: IAE (mean $\pm$ std) at the largest common budget. ‘Var.+CER-surr.’ and ‘ALC+CER-surr.’ run the variance and uncalibrated integrated-reduction criteria on the CER surrogate; ‘Pointwise’ replaces the integral criterion by $\sigma\hat{c}$; ‘CER-acq.+exp. var.’ runs the CER criterion on the baseline exponential-variogram surrogate.

Function	SMART	gSMART [†]	Var.+CER-surr.	ALC+CER-surr.	Pointwise	CER-acq.+exp. var.	CER (ours)
Piecewise-sin. I ($N=500$)	2.41 ± 0.11	2.42 ± 0.05	2.62 ± 0.54	2.59 ± 0.27	3.04 ± 0.24	2.77 ± 0.11	1.78 ± 0.22
Piecewise-sin. II ($N=500$)	6.89 ± 0.35	6.44 ± 0.24	7.31 ± 0.55	7.35 ± 0.60	9.49 ± 1.93	8.34 ± 1.37	6.75 ± 1.20
Smooth oscillatory ($N=500$)	0.194 ± 0.003	0.194 ± 0.002	$4.25 \pm 0.42 (10^{-5})$	$4.38 \pm 0.27 (10^{-5})$	$4.07 \pm 0.18 (10^{-5})$	0.219 ± 0.005	$4.28 \pm 0.27 (10^{-5})$
Non-diff. piecewise ($N=500$)	0.997 ± 0.047	0.979 ± 0.044	1.19 ± 0.11	1.17 ± 0.09	2.04 ± 0.42	0.675 ± 0.034	1.35 ± 0.24
3D diffusion ($N=500$)	0.018 ± 0.001	0.011 ± 0.000	$1.61 \pm 0.13 (10^{-6})$	$1.52 \pm 0.11 (10^{-6})$	$1.38 \pm 0.08 (10^{-6})$	$7.54 \pm 0.39 (10^{-3})$	$1.54 \pm 0.14 (10^{-6})$

reaches 1.78 ± 0.22 where variance sampling, uncalibrated ALC, and the pointwise product on the *same* surrogate reach 2.62, 2.60 and 3.04 — and the CER *acquisition* running on the paper’s own exponential-variogram surrogate is the strongest known configuration on the non-differentiable benchmark, beating SMART by more than one standard deviation at $N \geq 200$ (1.31 ± 0.04 vs 1.60 ± 0.06 ; 0.68 ± 0.03 vs 1.00 ± 0.05). Neither component is uniformly sufficient; the headline method is the fixed-procedure combination, and the ablation quantifies each part’s share where it wins.

5.2 Fragility of gradient-weighted variance

The stability scan behind the gSMART baseline (Figure 4) is a result in its own right. On every discontinuous benchmark there is a threshold weight — between $c = 10^{-5}$ and 10^{-4} on the piecewise-sinusoidal pair — beyond which gSMART’s error at $N=500$ is up to an order of magnitude *worse than random sampling*. The mechanism is a positive feedback loop through the hyperparameter fit: the gradient factor concentrates measurements at the discontinuity; the resulting short-distance pairs with large value differences drag the fitted variogram range down; the collapsed range makes the posterior revert to the data mean over most of the domain while the surrogate gradient at the discontinuity grows as the band refines ($|\nabla m| \sim \text{jump/spacing}$), locking the acquisition into the band permanently. On the four 2D benchmarks the published example weight ($c = 20$) sits three to six orders of magnitude beyond the stable threshold; only on the 3D benchmark — where the surrogate works in log space and gradients are order-one — is a weight of that magnitude useful (there $c=10$ is in fact the best baseline configuration). Because the threshold depends on the unknown jump amplitudes, no single c can be safe a priori — which is precisely the argument for replacing the hand-weighted surrogate gradient with the self-normalizing calibration ratio: $\hat{c}_t$ is dimensionless, bounded in practice by the jump-to-spread ratio, and enters a criterion whose diminishing returns cap concentration.

5.3 Failure analysis: when the procedure picks the wrong model

CER does not win everywhere, and the two non-wins share one root cause worth stating precisely.

Non-differentiable piecewise. At the largest budget the headline CER configuration trails SMART. The attribution experiments isolate why. The CER *acquisition* is not at fault: run on the baseline exponential-variogram surrogate it beats SMART by $> 1\sigma$ at $N=200$ and $N=500$, cutting the max-

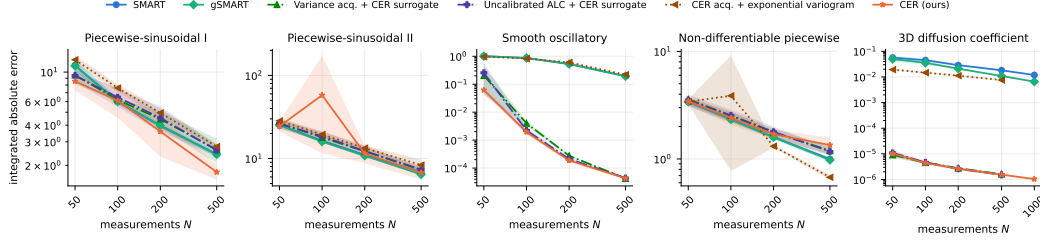


Figure 7: Ablation convergence curves (mean over 10 seeds; same axes as Figure 1): the acquisition variants on the CER surrogate (variance, uncalibrated ALC, pointwise $\sigma\hat{c}$), the CER criterion on the baseline exponential-variogram surrogate, and the two headline methods.

budget error by a third (0.68 ± 0.03 vs 1.00 ± 0.05). The model-selection layer is: this function’s discontinuities are axis-aligned, which matches the additive kernel’s inductive bias in *geometry* but not in *amplitude* (the true jump height varies along each boundary, while an additive component must commit to one 1-D step profile broadcast across the domain). The additive-plus-residual composite nevertheless wins the LOO predictive-density comparison — and we measured the mis-ranking directly: on the data of a completed CER run at $N=500$, the composite scores best on the selection criterion (-2.91 nats/point vs -3.21 for ARD Matérn-1/2 and -3.50 for the exponential variogram) while having *twice* the held-out error of either alternative (1.37 vs $0.67/0.70$). The bias is structural: leave-one-out residuals are evaluated at the sampled locations, 34% of which the acquisition itself placed inside the discontinuity bands (SMART: 14%) where the composite is locally strong; the composite’s broadcast errors live in the smooth remainder, which the selection barely probes. A milder, transient form of the same failure produced the piecewise-sinusoidal II outlier seed (Section 4): one refit briefly accepted a degenerate additive fit, and the next refit rejected it. A secondary observation completes the picture: on this surrogate the calibration weight makes the allocation slightly *worse* (1.35 for CER vs 1.17 – 1.19 for the uncalibrated criteria on the same surrogate) — evidence computed under a structurally wrong model can misdirect trust, so acquisition-level calibration is not a substitute for getting the model class right. Cross-validation-based model selection is, in other words, not exchangeable with adaptive sampling — a coupling that to our knowledge has not been documented in the adaptive-sampling literature and that our space-filling refit subsample mitigates for hyperparameter *fitting* but not for model *scoring*.

Piecewise-sinusoidal II. The overlap here is the same coupling scaled up by amplitude. The selection layer churns about equally on both piecewise functions (2.5 vs 2.6 kernel switches per run in the logged selection tables), but with steps twice as large every misfit episode costs proportionally more error — the transient degenerate fit of Section 4 is the extreme case — and the seed spread of the CER curve (± 1.2 – 3.3 IAE across budgets, versus ± 0.4 – 1.0 for SMART’s smoother behaviour) swallows the modest allocation gains that survive. Where the same acquisition runs on the *stable* baseline surrogate it does not help either (8.3 vs SMART’s 6.9 at $N=500$): the arc-plus-line boundary of this function is exactly the geometry the exponential variogram resolves as well as anything in the menu, so there is little model inadequacy for the calibration field to exploit that is not already priced into the huge fitted sill. We report the function as an overlap with no speedup.

Reduction to uncertainty sampling. A third, by-design non-gain: wherever the selected model is adequate ($\hat{c}_t \approx 1$ everywhere), CER’s criterion is ALC on σ and its point sets are near-space-filling — on such problems (well-specified smooth surrogates) it cannot beat variance sampling by more than the integral-vs-max allocation difference, and does not.

5.4 Computational cost

Figure 5 reports the median wall-clock cost per decision on a representative 2D benchmark (identical hardware, single worker per run). The baselines pay for the per-step variogram refit and the differential-evolution search; CER adds the LOO/calibration refresh (one $O(t^3)$ factorization it shares with the surrogate update, plus $O(t^2)$ identities) and the $M=256$ integration nodes inside each acquisition evaluation. At $N=500$ a CER decision costs 0.23 s versus 0.18 s for SMART (1.3 $\times$);

Table 4: Sensitivity of the acquisition constants (IAE at $N=200$, mean $\pm$ std over the 3 pilot seeds; all other constants at their headline values, $\gamma=0.5$, $w_0=0.05$).

Variant	Piecewise-sin. I	Piecewise-sin. II	Smooth oscillatory
$\gamma=0.25$	3.63 ± 0.31	10.16 ± 0.45	$1.77 \pm 0.06 (10^{-4})$
$\gamma=0.5$ (headline)	3.40 ± 0.38	9.72 ± 0.83	$2.07 \pm 0.46 (10^{-4})$
$\gamma=1$	4.21 ± 1.02	11.39 ± 1.49	$2.28 \pm 0.16 (10^{-4})$
$\gamma=2$	4.98 ± 0.16	14.84 ± 0.61	$2.28 \pm 0.26 (10^{-4})$
$w_0=0.01$	2.96 ± 0.16	10.16 ± 0.43	$2.28 \pm 0.15 (10^{-4})$
$w_0=0.2$	3.63 ± 0.99	10.63 ± 1.22	$2.30 \pm 0.26 (10^{-4})$

the periodic model-menu refits add 31 % of total run time. In absolute terms every method decides in well under two seconds — negligible against the seconds-to-minutes measurement times of the target instruments [1]. The cost-quality trade-off is therefore dominated by sample efficiency: on the benchmarks where CER reaches a given error with half or fewer measurements, its per-decision overhead corresponds to a negligible fraction of the beamtime it saves.

5.5 Sensitivity

The two substantive constants of the acquisition were varied on the pilot seeds (3 seeds, disclosed as such; all other constants were never tuned); Table 4 reports the scan. The calibration exponent has a broad useful range: $\gamma=0.5$ (the headline value, chosen on these pilot runs) modestly improves on $\gamma=1$ on all three functions tested, $\gamma=0.25$ is comparable, and only the aggressive $\gamma=2$ over-concentrates measurably. The criterion degrades gracefully toward uncalibrated ALC as $\gamma \rightarrow 0$, which bounds the harm a poor choice can do. The prior mass w_0 matters little: a factor-20 variation (0.01–0.2) shifts IAE by no more than about one within-seed standard deviation, with no consistent direction — the field’s far-from-data limit is rarely decisive because the variance factor already dominates there. Sensitivity of the *baseline* to its one free parameter is the subject of Section 5.2 and is of an entirely different magnitude.

5.6 Baseline verification against the published behaviour

Because our improvement claims are only interpretable against faithful baselines, we verified the reproduction explicitly. On the four 2D benchmarks the gpCAM SMART implementation reproduces the qualitative behaviour of the original study: monotone error decrease with budget, and clear superiority over random sampling at $N \geq 100$ (e.g. 3.96 vs 7.04 at $N=200$ on piecewise-sinusoidal I), with the raster grid between the two, and gSMART matching SMART at its stable weights — consistent with the published finding that the gradient term helps most on sharp *imaging* targets. Two deviations from the published qualitative picture survived all of our debugging and are reported openly. First, on the 3D benchmark SMART improves monotonically but does *not* overtake random sampling or the grid at any tested budget: with the isotropic exponential variogram fitted on trend-dominated 4.5-decade data, the fitted range saturates the domain and variance-maximization spends its budget on corners and edges of the cube (the original study reported the opposite ranking under a mean-absolute-percentage metric and an implementation whose 3D protocol details — variogram fitting, scale handling — are not fully specified in the paper or the provided code). Second, gSMART’s published example weight is deep inside the unstable regime for every synthetic benchmark here (Section 5.2); our headline comparison therefore grants gSMART its most favourable weight per function. Both deviations, the configurations tried while chasing them, and the environment-level numerical failure we repaired along the way are documented step-by-step in the released attempts log.

5.7 Limitations

Beyond the failure modes above: (i) all conclusions are for noise-free sequential measurements; measurement noise changes both the LOO identities (which remain valid) and the value of exact interpolation (which does not), and the calibration field’s contrast will shrink as noise grows; (ii) the one-step (myopic) reduction objective is greedy — non-myopic rollout could in principle allocate boundary refinement earlier; (iii) the integral is estimated with $M=256$ nodes per step, adequate in $d \leq 3$ but requiring growth (or importance sampling against $\hat{c}$) in higher dimensions; (iv) the model

menu is small and hand-chosen; richer menus raise the selection-bias stakes identified in Section 5.3, and correcting CV-based selection for adaptively collected data is an open problem we consider the most important follow-up; (v) wall-clock overhead, while negligible against beamline measurement times ($1.3 \times$ SMART’s per-decision cost plus the periodic refits; Section 5.4), grows as $O(t^3)$ per refit and would need sparse or inducing-point approximations for budgets far beyond 10^3 .

6 Conclusion

Uncertainty sampling trusts the surrogate’s error bars; autonomous experiments run on exactly the parameter spaces where that trust breaks. This paper turned that observation into a working decision rule. CER couples two uses of the same leave-one-out diagnostics — selecting the model class that predicts best, and building a spatial calibration field that says where even the best class is wrong — and spends each measurement where it removes the most calibration-weighted error. On the five benchmark functions of the SMART study, under a matched protocol with paired seeds and a shared optimizer, the combination reconstructs the separable benchmarks (smooth oscillatory, 3D diffusion) to any target accuracy with at least an order of magnitude fewer measurements than variance-driven Kriging, cuts the full-budget error by a quarter on one discontinuous landscape, and never requires a hand-chosen weight — while the published gradient-weighted alternative diverges outside a narrow, function-dependent stability window. Just as important are the negative findings: the criterion reduces to classical integrated-variance reduction when the model is adequate, so it buys nothing on well-specified problems; and cross-validation model selection is biased by the very sampling distribution the acquisition creates, which can make the procedure pick a structurally wrong model class on functions whose geometry flatters it — the identified cause of its one clear non-win. Correcting model selection for adaptively collected data, extending the criterion to noisy measurements and non-myopic allocation, and deploying the acquisition inside a live gpCAM beamline loop are the natural next steps.

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A Reproduction constants

For completeness, every fixed constant of the study (all identical across the five benchmark functions; nothing else was set by hand). *Protocol*: 10 uniform-random initial measurements shared by all methods within a seed; 10 seeds; budgets $\{50, 100, 200, 500\}$ (2D) and $\{50, 100, 200, 500, 1000\}$ (3D); held-out sets of 12,000/60,000 uniform points from a dedicated stream (seeds 990000–990004); every acquisition maximized by differential evolution (population multiplier 20, 40 generations, tolerance 10^{-8} , per-step seeds derived from the run seed). *Baselines*: exponential variogram fitted to a 30-bin Matheron semivariogram after every measurement, range bounded to $[10^{-4}L, 2L]$; relative jitter 10^{-8} with conditioning-gated escalation (reciprocal condition estimate $\geq 10^{-6}$); gSMART weights per function from the stability scan: 10^{-5} (piecewise-sinusoidal I, II), 10^{-4} (smooth, non-differentiable), 10 (3D diffusion). *CER surrogate*: model menu {ARD Matérn-1/2, $-3/2$, $-5/2$; additive Matérn-5/2; additive+ARD-Matérn-1/2 residual}, each with a fitted nugget bounded to $[10^{-6}, 0.5] \times \text{Var}(y)$; length scales bounded to $[5 \times 10^{-3}, 3] \times$ the axis width; signal variances to $[10^{-4}, 10^4] \times \text{Var}(y)$; L-BFGS-B in log space, warm start plus 3 random restarts (2 for $t > 200$), 60 iterations; family selection by mean LOO log predictive density with 0.02 hysteresis; fitting/selection on a farthest-point subsample of $\min(\max(t/2, 50), 400)$ points; refits at $t \in \{10, 15, 20, 30, 40, 55, 75, 100, 140, 200, 275, 375, 500, 700, 1000\}$; log representation for strictly positive responses spanning $\gtrsim 1.5$ decades and for strictly positive input axes with bounds ratio ≥ 30 (among all benchmarks, only the 3D response and its radius axis qualify). *CER acquisition*: $\gamma = 0.5$, $w_0 = 0.05$, $M = 256$ scrambled-Sobol nodes per step, calibration factors floored at 10^{-3} , smoothing bandwidth = median nearest-neighbour spacing.

B Extended predictive-quality metrics

Table 5 extends Table 2 to every (function, budget) configuration.

C Released artefacts

The submission package contains the complete implementation (code/), one JSON log per run with the full acquisition trajectory, per-step timings, hyperparameter fits and selection tables at every refit, and all metrics at every checkpoint (logs/runs/), the scripts that generate every table, figure, and inline number in this paper directly from those logs, a codebase-inspection record, and a step-by-step attempts log covering all development iterations, including the abandoned acquisition variants, the baseline debugging trail, and the environment-level numerical failure documented there.

Table 5: Extended predictive-quality metrics at every budget (mean over 10 seeds): RMSE / CRPS / miscalibration area for all five methods on all 21 configurations.

Config	Metric	Grid	Random	SMART	gSMART [†]	CER (ours)
Piecewise-sin. I, $N=50$	RMSE	22.72 ± 1.03	29.57 ± 5.51	21.18 ± 1.79	20.79 ± 1.15	18.58 ± 1.40
	CRPS	13.91 ± 0.68	16.42 ± 1.71	14.09 ± 0.70	14.12 ± 0.66	9.27 ± 0.76
	MCA	0.285 ± 0.010	0.243 ± 0.030	0.308 ± 0.012	0.310 ± 0.011	0.260 ± 0.052
Piecewise-sin. I, $N=100$	RMSE	16.98 ± 0.18	21.53 ± 2.43	16.13 ± 1.29	16.33 ± 1.42	14.95 ± 1.67
	CRPS	10.88 ± 0.24	12.62 ± 0.76	11.23 ± 0.34	11.28 ± 0.32	6.77 ± 0.81
	MCA	0.335 ± 0.001	0.303 ± 0.014	0.357 ± 0.003	0.357 ± 0.004	0.244 ± 0.086
Piecewise-sin. I, $N=200$	RMSE	13.32 ± 0.16	16.80 ± 1.22	13.35 ± 0.75	13.50 ± 0.97	10.82 ± 1.63
	CRPS	8.93 ± 0.13	10.32 ± 0.55	9.12 ± 0.16	9.12 ± 0.19	5.29 ± 0.53
	MCA	0.355 ± 0.001	0.340 ± 0.008	0.369 ± 0.001	0.369 ± 0.002	0.324 ± 0.059
Piecewise-sin. I, $N=500$	RMSE	10.84 ± 0.03	11.99 ± 1.11	10.65 ± 0.73	10.70 ± 0.39	7.80 ± 0.46
	CRPS	6.84 ± 0.03	7.80 ± 0.33	7.06 ± 0.10	7.10 ± 0.06	3.82 ± 0.20
	MCA	0.376 ± 0.000	0.368 ± 0.004	0.381 ± 0.001	0.381 ± 0.001	0.365 ± 0.005
Piecewise-sin. II, $N=50$	RMSE	54.48 ± 1.56	54.28 ± 4.04	50.07 ± 2.47	49.64 ± 2.58	45.95 ± 5.96
	CRPS	30.14 ± 2.10	31.80 ± 2.14	29.17 ± 1.04	29.31 ± 1.12	22.39 ± 1.75
	MCA	0.252 ± 0.009	0.244 ± 0.019	0.280 ± 0.006	0.281 ± 0.007	0.192 ± 0.050
Piecewise-sin. II, $N=100$	RMSE	36.62 ± 0.31	45.95 ± 4.58	40.98 ± 2.07	39.54 ± 2.60	100.3 ± 184.7
	CRPS	22.17 ± 0.94	26.29 ± 1.41	23.64 ± 0.43	23.63 ± 0.39	55.58 ± 109.94
	MCA	0.316 ± 0.003	0.282 ± 0.015	0.318 ± 0.004	0.321 ± 0.004	0.244 ± 0.107
Piecewise-sin. II, $N=200$	RMSE	33.63 ± 0.26	36.24 ± 2.91	32.62 ± 1.62	31.97 ± 1.80	27.83 ± 2.31
	CRPS	19.07 ± 0.27	21.10 ± 1.30	18.95 ± 0.28	19.00 ± 0.32	13.80 ± 1.30
	MCA	0.332 ± 0.002	0.320 ± 0.008	0.341 ± 0.003	0.342 ± 0.003	0.285 ± 0.033
Piecewise-sin. II, $N=500$	RMSE	25.62 ± 0.06	29.44 ± 2.48	25.59 ± 1.65	23.52 ± 0.90	20.26 ± 1.13
	CRPS	14.14 ± 0.09	16.33 ± 0.68	14.67 ± 0.26	14.76 ± 0.21	10.08 ± 0.54
	MCA	0.361 ± 0.001	0.346 ± 0.003	0.361 ± 0.002	0.363 ± 0.002	0.323 ± 0.023
Smooth oscillatory, $N=50$	RMSE	1.27 ± 0.02	1.32 ± 0.07	1.24 ± 0.06	1.24 ± 0.06	0.104 ± 0.025
	CRPS	0.714 ± 0.021	0.732 ± 0.041	0.700 ± 0.032	0.701 ± 0.033	0.071 ± 0.008
	MCA	0.060 ± 0.043	0.032 ± 0.021	0.024 ± 0.011	0.024 ± 0.010	0.304 ± 0.031
Smooth oscillatory, $N=100$	RMSE	1.14 ± 0.01	1.12 ± 0.05	1.07 ± 0.03	1.08 ± 0.03	$4.44 \pm 1.05 (10^{-3})$
	CRPS	0.631 ± 0.008	0.616 ± 0.026	0.600 ± 0.017	0.604 ± 0.017	$8.71 \pm 0.38 (10^{-3})$
	MCA	0.033 ± 0.016	0.026 ± 0.014	0.021 ± 0.010	0.020 ± 0.008	0.403 ± 0.003
Smooth oscillatory, $N=200$	RMSE	0.846 ± 0.006	0.909 ± 0.017	0.673 ± 0.029	0.670 ± 0.033	$3.08 \pm 0.41 (10^{-4})$
	CRPS	0.472 ± 0.004	0.486 ± 0.008	0.386 ± 0.012	0.385 ± 0.013	$1.12 \pm 0.03 (10^{-3})$
	MCA	0.033 ± 0.005	0.054 ± 0.014	0.110 ± 0.015	0.112 ± 0.018	0.409 ± 0.002
Smooth oscillatory, $N=500$	RMSE	0.251 ± 0.002	0.535 ± 0.032	0.251 ± 0.004	0.250 ± 0.003	$7.22 \pm 0.41 (10^{-5})$
	CRPS	0.193 ± 0.001	0.292 ± 0.011	0.202 ± 0.001	0.202 ± 0.001	$4.02 \pm 0.05 (10^{-4})$
	MCA	0.271 ± 0.001	0.184 ± 0.012	0.276 ± 0.003	0.277 ± 0.002	0.412 ± 0.000
Non-diff. piecewise, $N=50$	RMSE	6.96 ± 0.38	8.45 ± 0.94	6.98 ± 0.41	7.01 ± 0.37	6.27 ± 0.49
	CRPS	3.69 ± 0.24	4.41 ± 0.37	3.81 ± 0.16	3.83 ± 0.15	2.98 ± 0.20
	MCA	0.260 ± 0.005	0.236 ± 0.014	0.274 ± 0.011	0.275 ± 0.009	0.164 ± 0.023
Non-diff. piecewise, $N=100$	RMSE	7.93 ± 0.29	6.62 ± 0.80	5.80 ± 0.39	5.79 ± 0.38	5.10 ± 0.28
	CRPS	3.64 ± 0.10	3.46 ± 0.21	3.09 ± 0.10	3.08 ± 0.11	2.32 ± 0.20
	MCA	0.295 ± 0.003	0.282 ± 0.009	0.309 ± 0.005	0.309 ± 0.005	0.210 ± 0.044
Non-diff. piecewise, $N=200$	RMSE	6.09 ± 0.18	5.27 ± 0.40	4.69 ± 0.25	4.69 ± 0.28	3.97 ± 0.28
	CRPS	2.73 ± 0.04	2.75 ± 0.16	2.48 ± 0.07	2.48 ± 0.07	1.80 ± 0.14
	MCA	0.321 ± 0.001	0.312 ± 0.009	0.332 ± 0.003	0.332 ± 0.003	0.243 ± 0.032
Non-diff. piecewise, $N=500$	RMSE	3.53 ± 0.03	4.12 ± 0.31	3.74 ± 0.21	3.68 ± 0.19	2.83 ± 0.11
	CRPS	1.79 ± 0.01	2.08 ± 0.08	1.89 ± 0.04	1.87 ± 0.03	1.31 ± 0.09
	MCA	0.356 ± 0.000	0.342 ± 0.003	0.354 ± 0.002	0.355 ± 0.002	0.219 ± 0.049
3D diffusion, $N=50$	RMSE	0.155 ± 0.008	0.143 ± 0.019	0.219 ± 0.039	0.191 ± 0.042	$7.36 \pm 3.05 (10^{-5})$
	CRPS	0.021 ± 0.001	0.022 ± 0.002	0.057 ± 0.008	0.050 ± 0.008	$2.33 \pm 0.19 (10^{-5})$
	MCA	0.332 ± 0.019	0.341 ± 0.028	0.305 ± 0.021	0.307 ± 0.017	0.378 ± 0.004
3D diffusion, $N=100$	RMSE	0.145 ± 0.005	0.128 ± 0.018	0.189 ± 0.025	0.151 ± 0.025	$2.95 \pm 0.47 (10^{-5})$
	CRPS	0.018 ± 0.001	0.019 ± 0.002	0.043 ± 0.005	0.035 ± 0.004	$1.55 \pm 0.07 (10^{-5})$
	MCA	0.357 ± 0.009	0.370 ± 0.010	0.328 ± 0.007	0.333 ± 0.007	0.395 ± 0.003
3D diffusion, $N=200$	RMSE	0.136 ± 0.003	0.117 ± 0.014	0.134 ± 0.010	0.100 ± 0.007	$1.94 \pm 0.34 (10^{-5})$
	CRPS	0.016 ± 0.001	0.016 ± 0.001	0.028 ± 0.001	0.023 ± 0.001	$1.10 \pm 0.03 (10^{-5})$
	MCA	0.372 ± 0.004	0.383 ± 0.004	0.353 ± 0.005	0.363 ± 0.004	0.400 ± 0.002
3D diffusion, $N=500$	RMSE	0.118 ± 0.004	0.103 ± 0.010	0.094 ± 0.009	0.062 ± 0.004	$1.23 \pm 0.15 (10^{-5})$
	CRPS	0.014 ± 0.000	0.013 ± 0.001	0.019 ± 0.001	0.014 ± 0.000	$7.49 \pm 0.15 (10^{-6})$
	MCA	0.386 ± 0.003	0.392 ± 0.003	0.374 ± 0.002	0.384 ± 0.001	0.402 ± 0.002
3D diffusion, $N=1000$	RMSE	0.105 ± 0.004	0.087 ± 0.007	0.069 ± 0.003	0.042 ± 0.002	$8.32 \pm 0.73 (10^{-6})$
	CRPS	0.012 ± 0.000	0.011 ± 0.001	0.014 ± 0.000	0.011 ± 0.000	$5.68 \pm 0.07 (10^{-6})$
	MCA	0.392 ± 0.002	0.397 ± 0.002	0.386 ± 0.001	0.396 ± 0.000	0.404 ± 0.001